Satellite Identification of Damage in Beirut, Lebanon

29 September 2006

Example of Before and After Satellite Image of Bomb Attacks on Beirut

Version 1.0

Crisis Background Information

Following the 12 July 2006 abduction of two Israeli soldiers by Hezbollah, the Israeli Defense Force (IDF) launched a large scale military operation in Lebanon. Simultaneously, Hezbollah launched rocket attacks into Israel. This crisis resulted in hundreds of deaths and injuries on both sides, massive population displacement and extensive damage to civilian infrastructure. The United Nations Security Council Resolution 1701 (2006), calling for an immediate cessation of hostilities, was passed on 11 August 2006. A cease fire based on the creation of a buffer zone free of armed personnel other that UN and Lebanese forces consequentially came into effect on Monday 14 August 2006.

In support to the United Nations Office for the Coordination of Humanitarian Affairs (UN OCHA), UNOSAT activated its 24/7 Rapid Mapping service to quickly provide the international humanitarian community and local actors with detailed satellite map atlases and damage assessments of key affected villages.

Map Information

Damage assessment has been done using photo-interpretation of pre and post-crisis Ikonos satellite imagery recorded 17 June 2005 and 19 August 2006 respectively. The depiction and use of boundaries, geographic names and related data shown here are not warranted to be error-free nor do they imply official endorsement or acceptance by the United Nations. This map was produced by the United Nations Institute for Training and Research (UNITAR) Operational Satellite Applications Programme (UNOSAT). UNOSAT provides satellite imagery and related geographic information to UN humanitarian and development agencies and their implementing partners.

Damage Assessment Summary

On the adjacent subscene a total of 180 damage sites were identified. Of this total, the following 178 sites were either destroyed or severely damaged:

- 178 Buildings (general)
- 0 Towers / Antennas
- 0 High-rise apartment buildings
- 0 Bridges
- 0 Irrigation canals
- 0 Agricultural buildings

The remaining 2 impact sites were located as follows:

- 2 on roads
- 0 in cultivated fields
- 0 in uncultivated fields

An additional 718 buildings (all types) were identified as within the 100 meter blast radius of the 180 identified impact sites, and may be lightly to severely damaged.

NOTE: Sites classified as "Building (general)" may include residential, business, industrial or agricultural building types that could not be classified further with the available satellite imagery.

Map Legend

Site locations of identified damage

- Bridge
- Road
- High-rise apartment building
- Agricultural building
- Building (general)
- Field (cultivated)
- Irrigation canal
- Field (uncultivated)
- Industrial building
- Airport Runway
- 100m Blast Radius

Scale (overview map) 1:2,500 for A1 Print

Meters: 0 50 100 150 200

Feet: 0 100 200 300 400 500 600

Satellite image ..... Ikonos (1m)	Data Sources ..... GIST, UNOSAT
Pre Image Date ..... 17 June 2005	Datum ..... WGS 1984
Post Image Date ..... 19 August 2006	False Easting 500000.00
Image Copyright ..... European Space Imaging	False Northing 00.00000
Map Production ..... UNOSAT (29 Sept. 2006)	Central Meridian 39.00000
Projection ..... UTM Zone 36N	Scale_Factor 0.999600

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